



PROFESSIONAL SUMMARY

Hello , I'm Pavithra — a Computer Engineering graduate who loves building real-practical things that actually work. I mainly build AI-powered apps using RAG, LLMs, and vector search, and pair that with clean, responsive frontends in React and Tailwind CSS. I've built end-to-end projects so far, interned across AI/ML and web development, and I am passionate about learning through practical projects and continuously improving my technical skills.

KEY EXPERTISE

Python C++ SQL HTML/Tailwind CSS React.js Machine Learning MySQL Docker GenAI AWS LLMS
RAG Prompt Engineering Javascript

EDUCATION

SIES Graduate School of Technology

2022 - 2026

B.E. - Computer Engineering | CGPA: **7.93 / 10**

CHANGU KANA THAKUR SECONDARY & HIGHER SECONDARY VIDYALAYA

2022

12th | MSBSHSE | Percentage: **65.33 / 100**

ST.JOSEPH'S HIGH SCHOOL

2020

10th | MSBSHSE | Percentage: **80.40 / 100**

INTERNSHIPS

Oneture Technologies | IT Product & Services

01 Dec, 2025 - 31 Jan, 2026

AIML

Key Skills: Rust Python System Engineering C++ Artificial Software Testing

Developed a high-performance Supplier Profile Dashboard using React.js and optimized low-level networking protocols application in C++ and Rust to ensure efficient data handling.

Integrated pre-trained ML models into the platform for automated data classification, engineering scalable system pipelines that optimized model inference speeds and prepared the infrastructure for future AI/ML deployment.

Vault-Tech Security

01 Jan, 2025 - 31 Jan, 2025

Web Developer

Key Skills: HTML5 CSS Flexbox Javascript API Integration Backend Development Database Management

Worked on a real-world project Subscription Management System, where I built a responsive frontend (HTML, CSS, JS), developed backend features (authentication, subscription tracking, payments), managed databases, integrated APIs, and deployed a secure, user-friendly system.

IIT Bombay

09 Feb, 2024 - 09 Mar, 2024

Frontend Developer

Key Skills: Software Design & Research Collaboration AI Chatbot Integration HTML CSS Javascript

Developed a solution to make programming error messages and technical keywords more understandable for beginners, addressing the limitations of existing tools like Google Translate that provide complex or literal translations.

PROJECTS

SmartBuddy

Key Skills:

Python Data Structures & Algorithms Vector Search RAG LLMs Ollama KD-Tree
HNSW (Hierarchical Navigable Small World) Machine Learning

Project Link: <https://github.com/Pavithra29-eng/SmartBuddy>

SmartBuddy is a Private AI search engine and chatbot built from scratch in Python, featuring vector search, Brute Force, KD-Tree, HNSW, and local RAG-powered document Q&A using Ollama.

PrepPulse

Key Skills: Gemini 2.5 Flash Backend Database Frontend (Html, React.js, Tailwind CSS & SCSS, API integration)

Project Link: <https://preppulse-mu.vercel.app/>

PrepPulse is a smart, free web application built to help job seekers prepare for interviews and upload our resume, paste the job description, and get a personalized interview prep plan plus a tailored resume — powered by AI.

C++ChatCore

Key Skills: Language: C++ Networking: Winsock2 Database: SQLite3

Project Link: <https://github.com/Pavithra29-eng/Cpp-Winsock-Chat-Application>

I wanted to understand how chat apps like WhatsApp actually work under the hood, so I challenged myself to build one from scratch.

This project is a multi-user chat server built using raw C++ and basic networking (Winsock). It lets people connect, register, and send messages instantly.

CodeMitra

Key Skills: Python Streamlit Regex Generative AI LLMs NLP Multilingual AI Streamlit Community Cloud

Project Link: <https://codemitra.streamlit.app/>

CodeMitra is a multilingual programming error explanation application designed to help beginner programmers, especially learners from rural and non-English-speaking backgrounds, understand programming errors in their native languages.

SEMINARS / TRAININGS / WORKSHOPS

Generative AI: Prompt Engineering Basics Institute Name: IBM (Coursera)

15 Jul, 2026 - 15 Jul, 2026

Key Skills:

Prompt Patterns ChatGPT Generative AI Prompt Engineering

Learned the fundamentals of crafting effective prompts for generative AI models.

Learned core prompt engineering techniques — zero-shot and few-shot prompting, prompt refinement, and best practices for guiding generative AI models like ChatGPT and other LLMs to produce accurate, relevant outputs.

SQL USING AI WORKSHOP

31 Aug, 2025 - 31 Aug, 2025

Institute Name: AI For Techies

Key Skills: SQL

I completed a 3-hour SQL Using AI Workshop by AI for Techies. I learned how to extract and analyze data, use SQL to query and preprocess datasets, and automate simple data tasks. This workshop improved my data skills and gave me knowledge useful for roles like a data analyst or database administrator.

Developer and Technology

15 Dec, 2024 - 17 Dec, 2024

Institute Name: Accenture

Key Skills:

Software Development Methodologies: Waterfall and Agile Software Development Lifecycle (SDLC) Code Debugging
Software Testing Lifecycle (STLC)

Gained a solid understanding of foundational software engineering concepts, covering the Software Development Lifecycle (SDLC) and methodologies like Waterfall and Agile. Developed core problem-solving skills through algorithmic thinking, code debugging, and the Software Testing Lifecycle (STLC).

WEB LINKS

- o Github - <https://github.com/Pavithra29-eng>